Results

External regulation

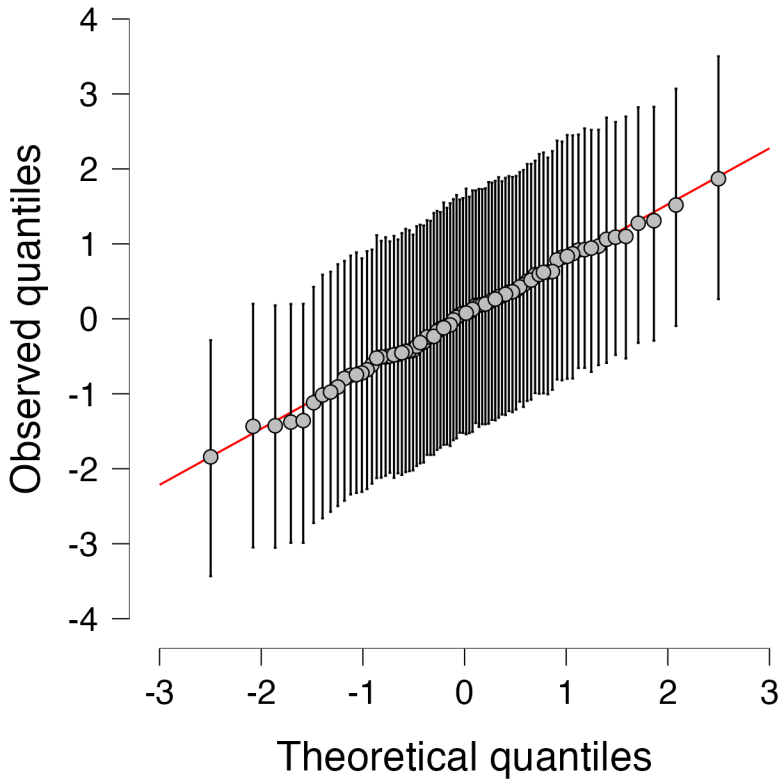
Bayesian Repeated Measures ANOVA

Model Comparison

Models	P(M)	P(M|data)	BF_M	BF_{10}	error %
Null model (incl. subject and random slopes)	0.200	0.038	0.159	1.000	
Time + group	0.200	0.445	3.204	11.602	6.087
Time	0.200	0.379	2.439	9.882	1.122
Time + group + Time ✻ group	0.200	0.082	0.359	2.149	3.480
group	0.200	0.056	0.236	1.454	20.627

Note. All models include subject, and random slopes for all repeated measures factors.

Model Averaged Q-Q Plot



Post Hoc Tests

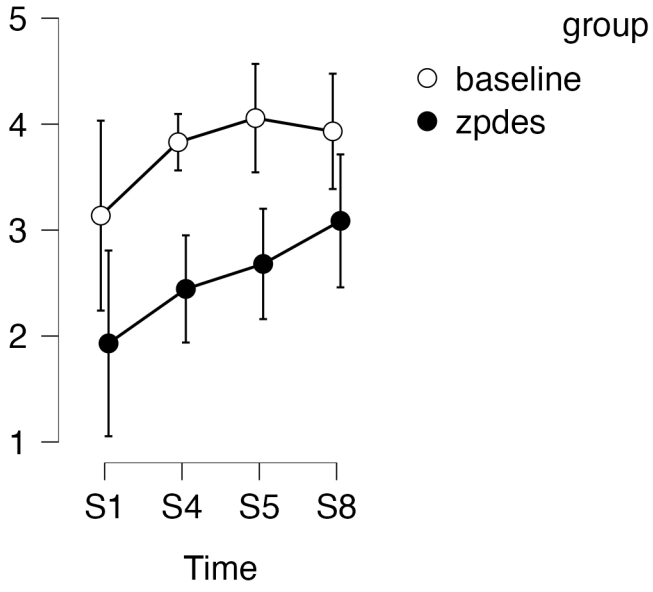
Post Hoc Comparisons - group

		Prior Odds	Posterior Odds	$BF_{10, U}$	error %
baseline	zpdes	1.000	13.722	13.722	2.403×10^{-7}

Note. The posterior odds have been corrected for multiple testing by fixing to 0.5 the prior probability that the null hypothesis holds across all comparisons (Westfall, Johnson, & Utts, 1997). Individual comparisons are based on the default t-test with a Cauchy (0, r = 1/sqrt(2)) prior. The "U" in the Bayes factor denotes that it is uncorrected.

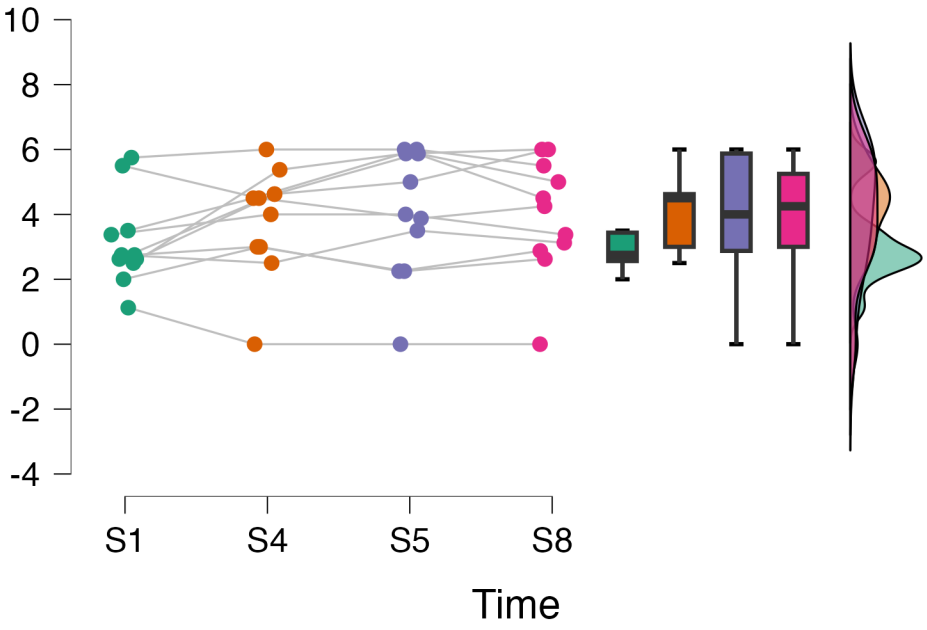
Descriptives

Descriptives plots

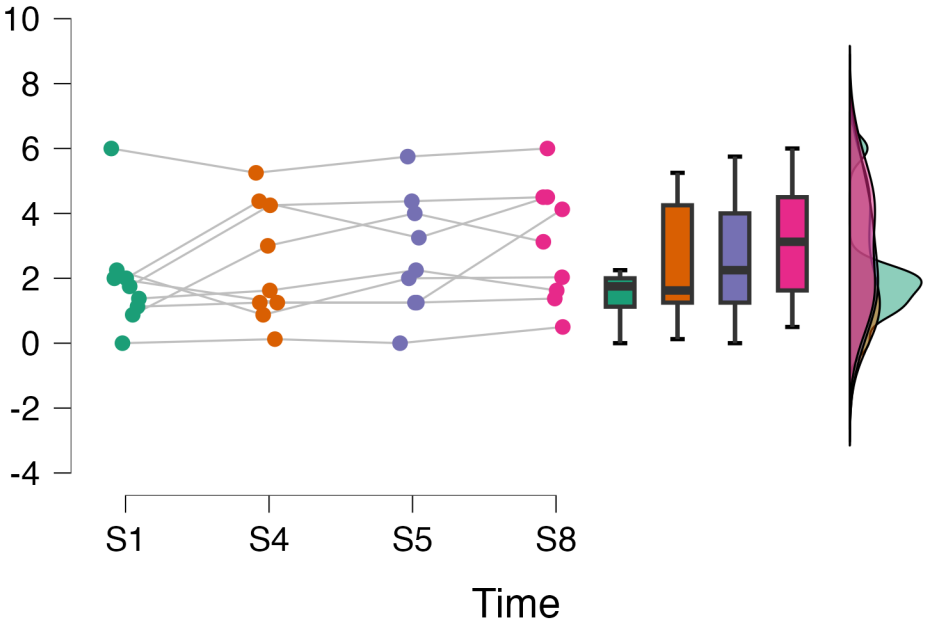


Raincloud plots

Dependent: group: baseline



Dependent: group: zpdes



Repeated Measures ANOVA

Within Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
RM Factor 1	11.020[a]	3[a]	3.673[a]	4.843[a]	0.005[a]
RM Factor 1 ✻ group	0.947[a]	3[a]	0.316[a]	0.416[a]	0.742[a]
Residuals	40.960	54	0.759		

Note. Type III Sum of Squares

[a] Mauchly's test of sphericity indicates that the assumption of sphericity is violated (p < .05).

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
group	28.657	1	28.657	2.910	0.105
Residuals	177.279	18	9.849		

Note. Type III Sum of Squares

Descriptives

Descriptives

RM Factor 1	group	N	Mean	SD	SE	Coefficient of variation
S1	baseline	11	3.136	1.385	0.417	0.441
	zpdes	9	1.931	1.677	0.559	0.869
S4	baseline	11	3.830	1.646	0.496	0.430
	zpdes	9	2.444	1.821	0.607	0.745
S5	baseline	11	4.057	1.959	0.591	0.483
	zpdes	9	2.681	1.813	0.604	0.676
S8	baseline	11	3.932	1.784	0.538	0.454
	zpdes	9	3.087	1.818	0.606	0.589

Assumption Checks

Test for Equality of Variances (Levene's)

	F	df1	df2	p
value_1	2.883×10^{-4}	1	18	0.987
value_4	0.707	1	18	0.411
value_5	0.017	1	18	0.896
value_8	0.082	1	18	0.778

Results

Amotivation

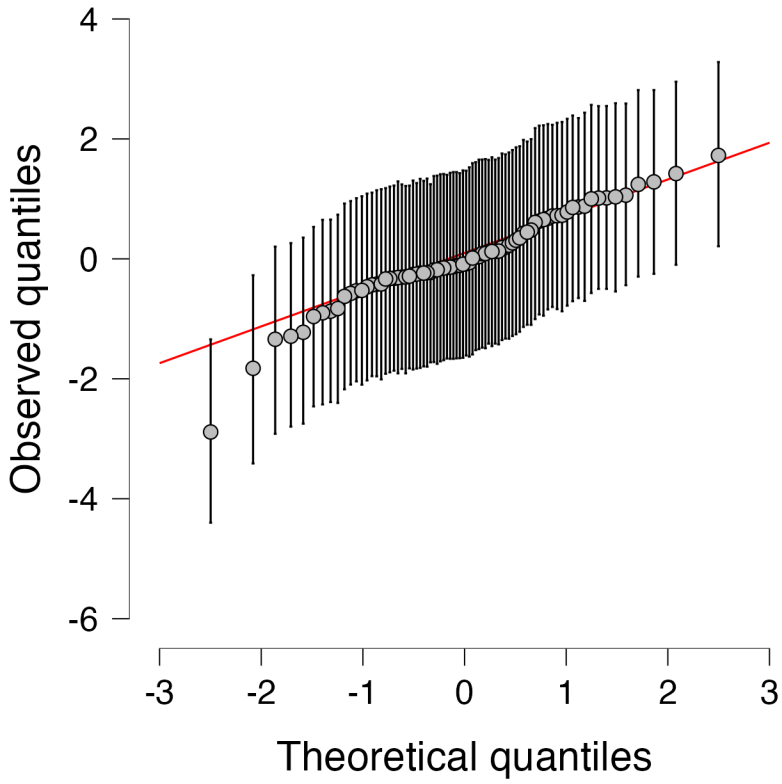
Bayesian Repeated Measures ANOVA

Model Comparison

Models	P(M)	P(M|data)	BF_M	BF_{10}	error %
Null model (incl. subject and random slopes)	0.200	0.422	2.922	1.000	
group	0.200	0.331	1.983	0.785	2.479
Time	0.200	0.129	0.591	0.305	0.353
Time + group	0.200	0.101	0.449	0.239	2.493
Time + group + Time ✻ group	0.200	0.017	0.068	0.040	4.111

Note. All models include subject, and random slopes for all repeated measures factors.

Model Averaged Q-Q Plot



Post Hoc Tests

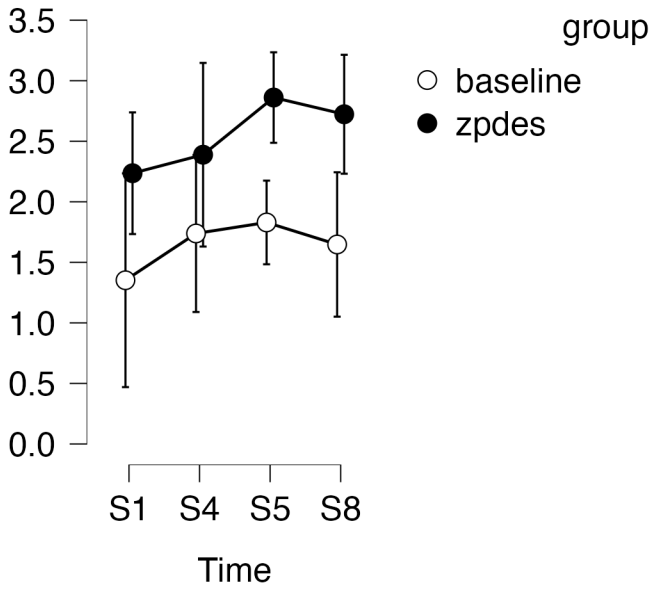
Post Hoc Comparisons - group

		Prior Odds	Posterior Odds	$BF_{10, U}$	error %
baseline	zpdes	1.000	1.807	1.807	0.010

Note. The posterior odds have been corrected for multiple testing by fixing to 0.5 the prior probability that the null hypothesis holds across all comparisons (Westfall, Johnson, & Utts, 1997). Individual comparisons are based on the default t-test with a Cauchy (0, r = 1/sqrt(2)) prior. The "U" in the Bayes factor denotes that it is uncorrected.

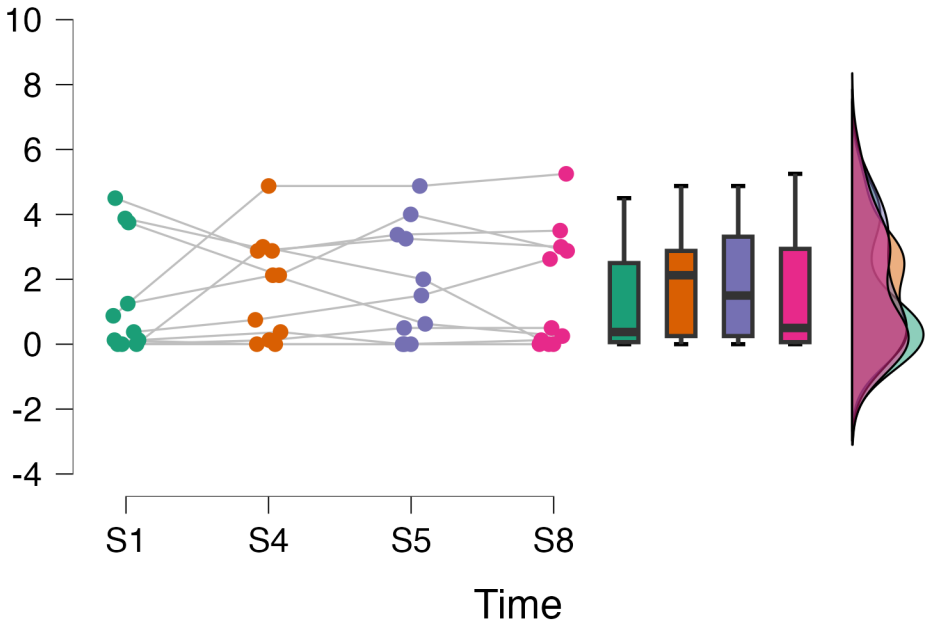
Descriptives

Descriptives plots

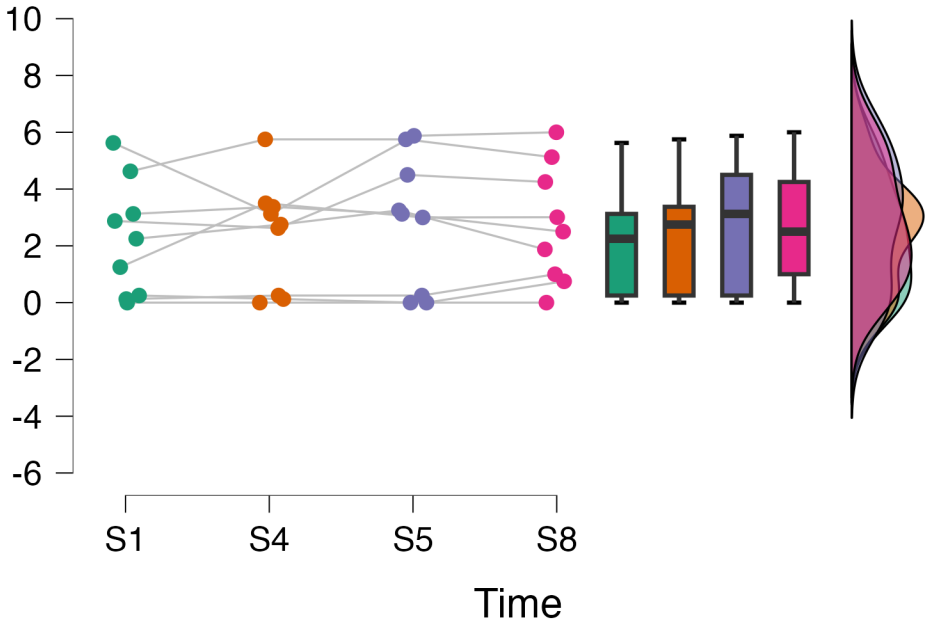


Raincloud plots

Dependent: group: baseline



Dependent: group: zpdes



Repeated Measures ANOVA

Within Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
RM Factor 1	3.213[a]	3[a]	1.071[a]	1.442[a]	0.241[a]
RM Factor 1 ✻ group	0.546[a]	3[a]	0.182[a]	0.245[a]	0.865[a]
Residuals	40.100	54	0.743		

Note. Type III Sum of Squares

[a] Mauchly's test of sphericity indicates that the assumption of sphericity is violated (p < .05).

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
group	16.406	1	16.406	1.321	0.265
Residuals	223.544	18	12.419		

Note. Type III Sum of Squares

Descriptives

Descriptives

RM Factor 1	group	N	Mean	SD	SE	Coefficient of variation
S1	baseline	11	1.352	1.781	0.537	1.317
	zpdes	9	2.236	2.024	0.675	0.905
S4	baseline	11	1.739	1.606	0.484	0.924
	zpdes	9	2.389	1.925	0.642	0.806
S5	baseline	11	1.830	1.783	0.538	0.975
	zpdes	9	2.861	2.333	0.778	0.815
S8	baseline	11	1.648	1.856	0.559	1.126
	zpdes	9	2.723	2.061	0.687	0.757

Assumption Checks

Test for Equality of Variances (Levene's)

	F	df1	df2	p
value_1	0.134	1	18	0.719
value_4	0.147	1	18	0.706
value_5	0.518	1	18	0.481
value_8	0.004	1	18	0.948

Results

Identified regulation

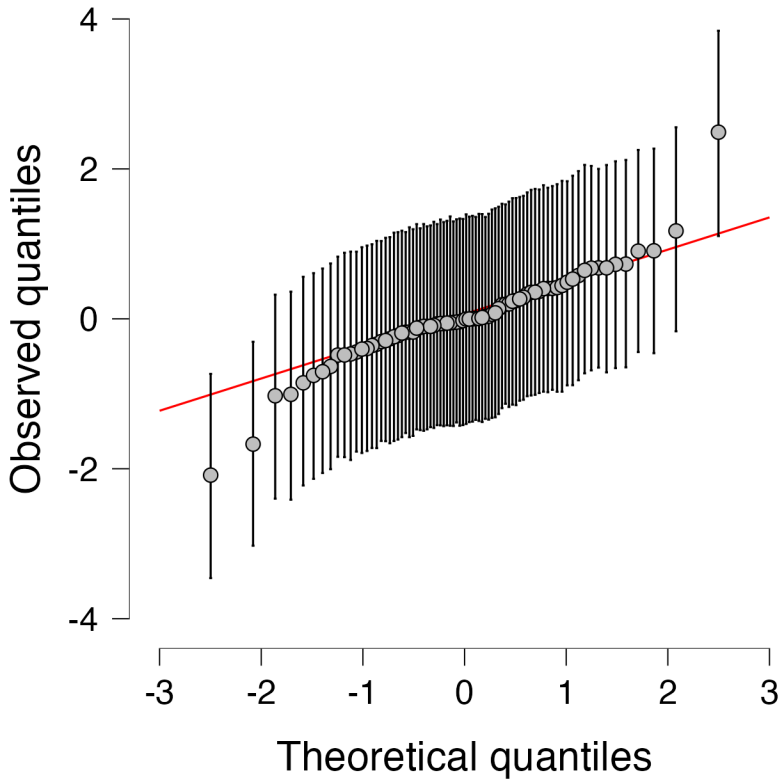
Bayesian Repeated Measures ANOVA

Model Comparison

Models	P(M)	P(M|data)	BF_M	BF_{10}	error %
Null model (incl. subject and random slopes)	0.200	0.238	1.247	1.000	
Time	0.200	0.351	2.161	1.475	1.446
Time + group	0.200	0.224	1.153	0.941	2.903
group	0.200	0.153	0.725	0.646	2.107
Time + group + Time ✻ group	0.200	0.034	0.142	0.144	4.695

Note. All models include subject, and random slopes for all repeated measures factors.

Model Averaged Q-Q Plot



Post Hoc Tests

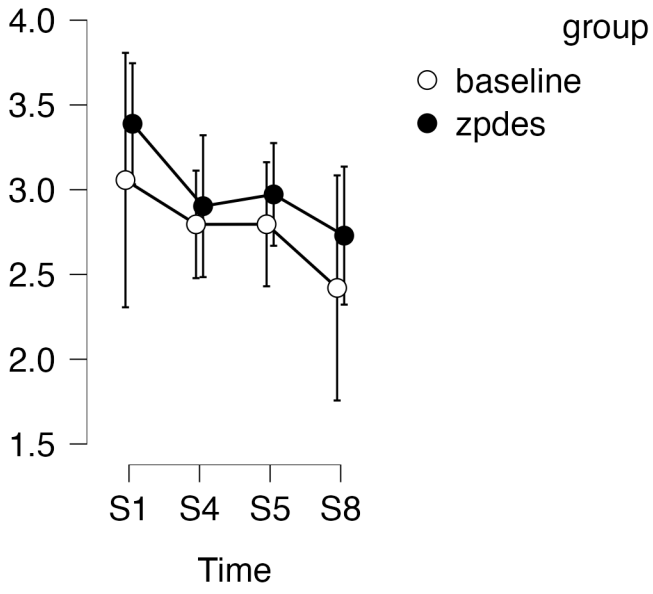
Post Hoc Comparisons - group

		Prior Odds	Posterior Odds	$BF_{10, U}$	error %
baseline	zpdes	1.000	0.271	0.271	0.017

Note. The posterior odds have been corrected for multiple testing by fixing to 0.5 the prior probability that the null hypothesis holds across all comparisons (Westfall, Johnson, & Utts, 1997). Individual comparisons are based on the default t-test with a Cauchy (0, r = 1/sqrt(2)) prior. The "U" in the Bayes factor denotes that it is uncorrected.

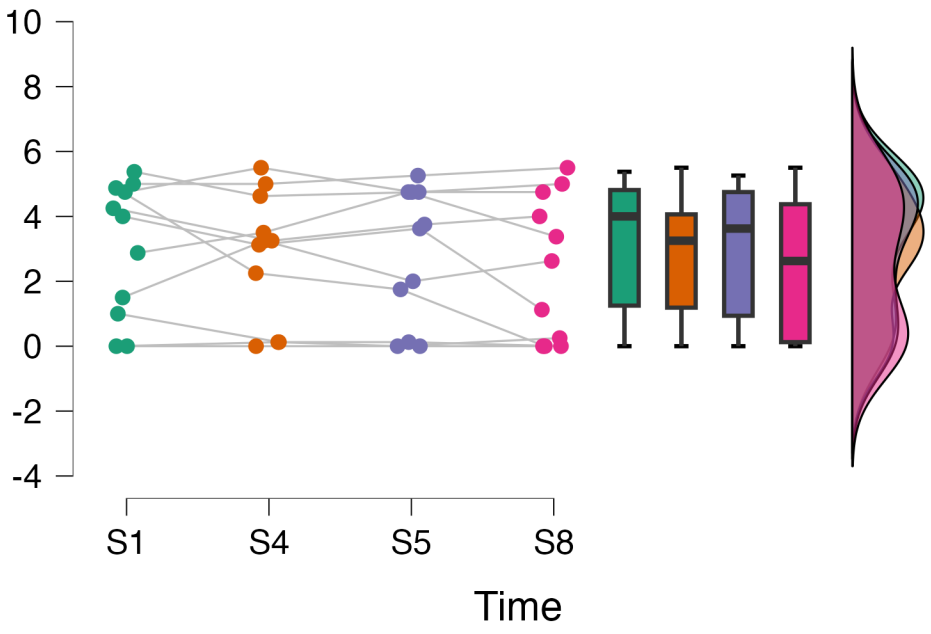
Descriptives

Descriptives plots

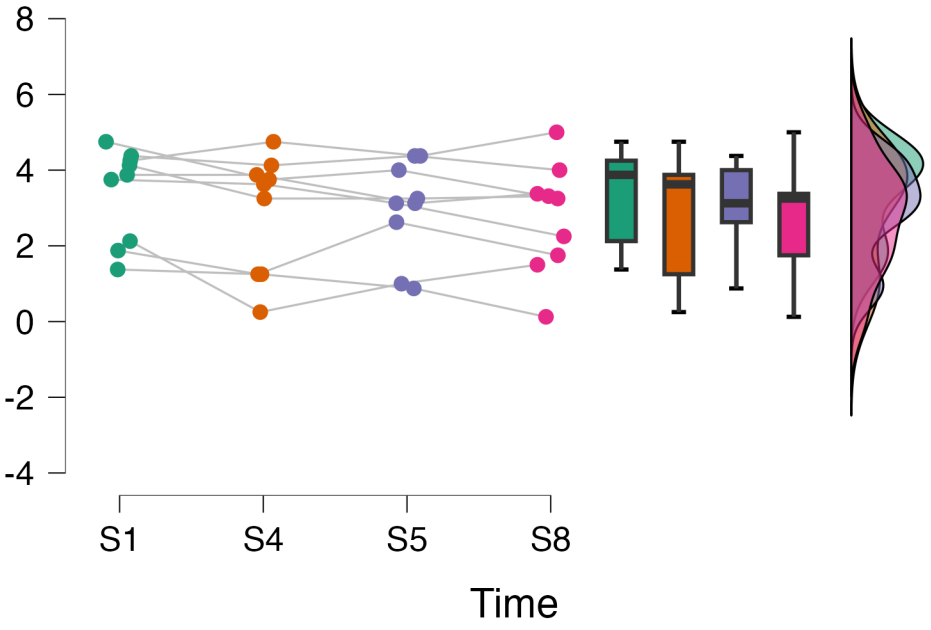


Raincloud plots

Dependent: group: baseline



Dependent: group: zpdes



Repeated Measures ANOVA

Within Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
RM Factor 1	4.190[a]	3[a]	1.397[a]	2.871[a]	0.045[a]
RM Factor 1 ✻ group	0.171[a]	3[a]	0.057[a]	0.117[a]	0.950[a]
Residuals	26.268	54	0.486		

Note. Type III Sum of Squares

[a] Mauchly's test of sphericity indicates that the assumption of sphericity is violated (p < .05).

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
group	1.057	1	1.057	0.090	0.768
Residuals	211.182	18	11.732		

Note. Type III Sum of Squares

Descriptives

Descriptives

RM Factor 1	group	N	Mean	SD	SE	Coefficient of variation
S1	baseline	11	3.057	2.074	0.625	0.678
	zpdes	9	3.389	1.246	0.415	0.368
S4	baseline	11	2.795	1.973	0.595	0.706
	zpdes	9	2.903	1.570	0.523	0.541
S5	baseline	11	2.796	2.088	0.630	0.747
	zpdes	9	2.972	1.300	0.433	0.437
S8	baseline	11	2.420	2.213	0.667	0.914
	zpdes	9	2.729	1.470	0.490	0.539

Assumption Checks

Test for Equality of Variances (Levene's)

	F	df1	df2	p
value_1	5.081	1	18	0.037
value_4	0.374	1	18	0.549
value_5	5.763	1	18	0.027
value_8	4.466	1	18	0.049

Results

Intrinsic motivation

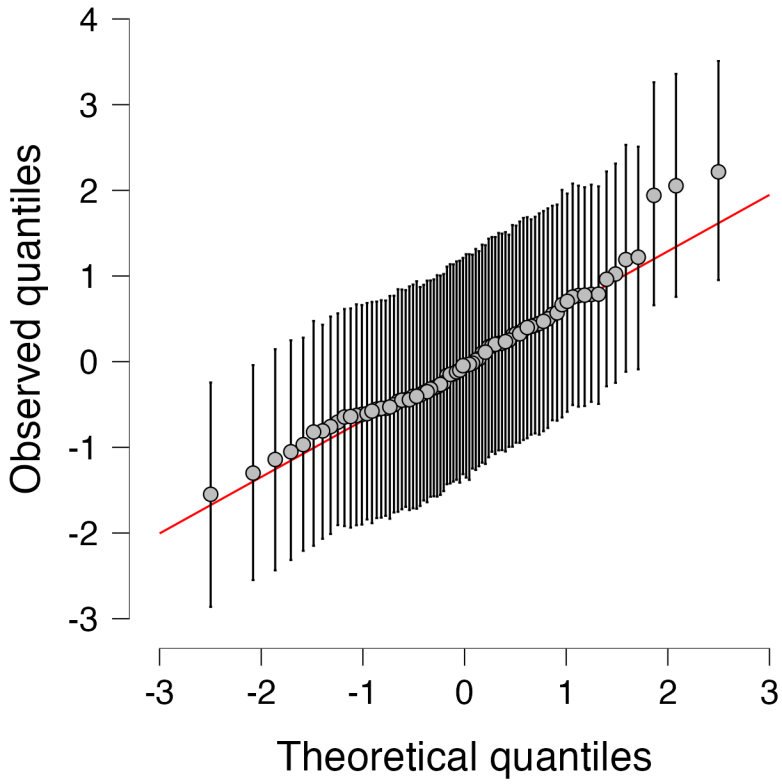
Bayesian Repeated Measures ANOVA

Model Comparison

Models	P(M)	P(M|data)	BF_M	BF_{10}	error %
Null model (incl. subject and random slopes)	0.200	0.173	0.834	1.000	
Time	0.200	0.442	3.173	2.564	0.475
Time + group	0.200	0.254	1.364	1.474	2.401
group	0.200	0.096	0.423	0.555	1.729
Time + group + Time ✻ group	0.200	0.035	0.146	0.204	3.919

Note. All models include subject, and random slopes for all repeated measures factors.

Model Averaged Q-Q Plot



Post Hoc Tests

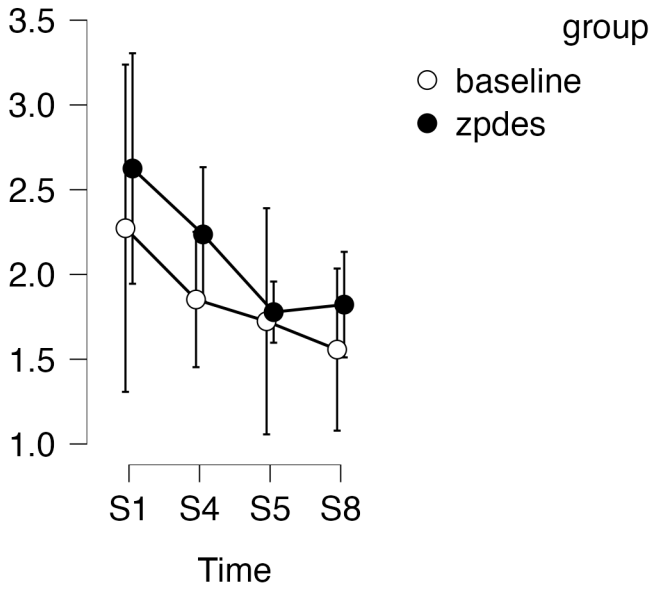
Post Hoc Comparisons - group

		Prior Odds	Posterior Odds	$BF_{10, U}$	error %
baseline	zpdes	1.000	0.301	0.301	0.016

Note. The posterior odds have been corrected for multiple testing by fixing to 0.5 the prior probability that the null hypothesis holds across all comparisons (Westfall, Johnson, & Utts, 1997). Individual comparisons are based on the default t-test with a Cauchy (0, r = 1/sqrt(2)) prior. The "U" in the Bayes factor denotes that it is uncorrected.

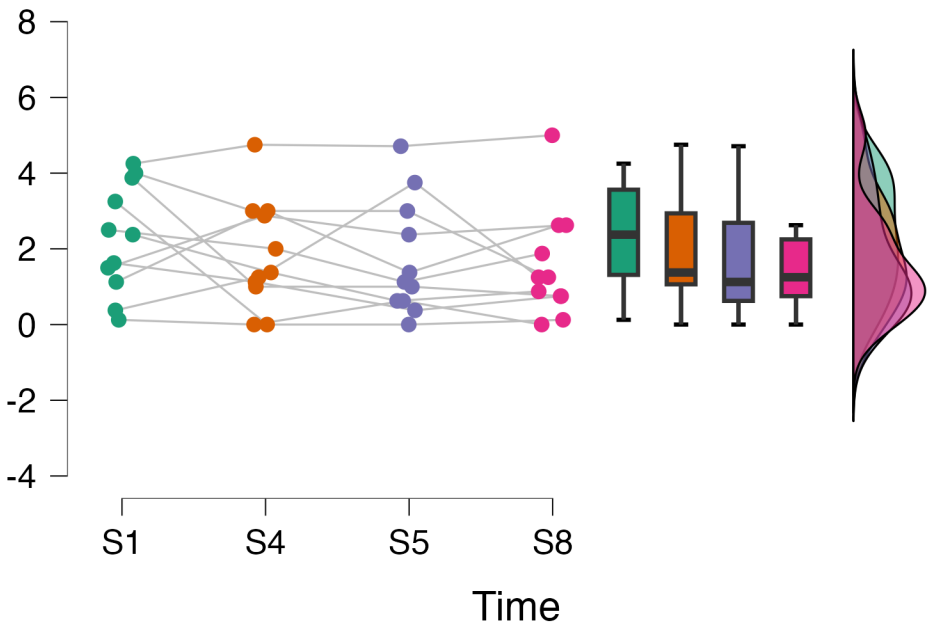
Descriptives

Descriptives plots

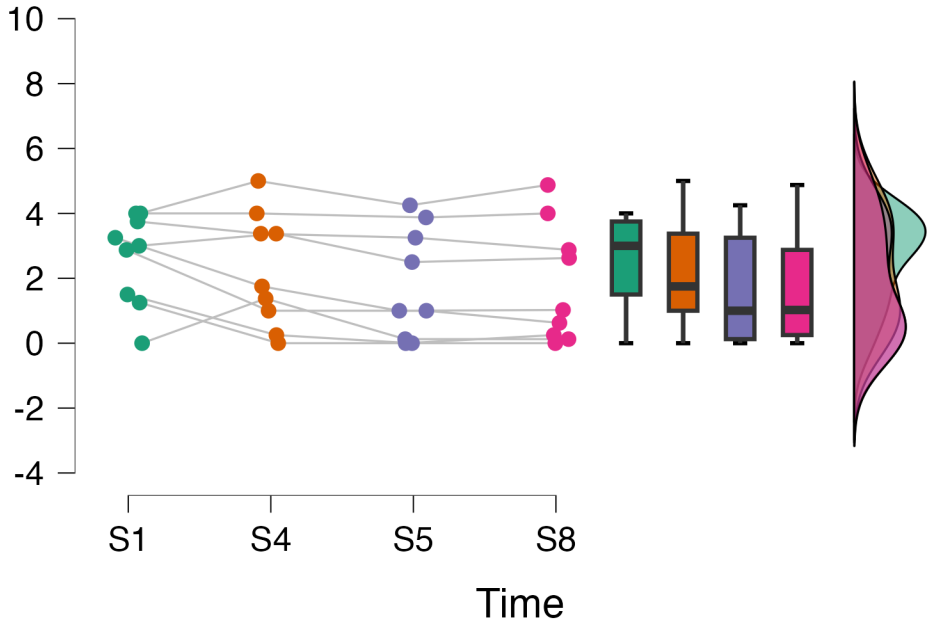


Raincloud plots

Dependent: group: baseline



Dependent: group: zpdes



Repeated Measures ANOVA

Within Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
RM Factor 1	7.146[a]	3[a]	2.382[a]	3.483[a]	0.022[a]
RM Factor 1 ✻ group	0.328[a]	3[a]	0.109[a]	0.160[a]	0.923[a]
Residuals	36.930	54	0.684		

Note. Type III Sum of Squares

[a] Mauchly's test of sphericity indicates that the assumption of sphericity is violated ($p < .05$).

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
group	1.378	1	1.378	0.178	0.678
Residuals	139.364	18	7.742		

Note. Type III Sum of Squares

Descriptives

Descriptives

RM Factor 1	group	N	Mean	SD	SE	Coefficient of variation
S1	baseline	11	2.273	1.450	0.437	0.638
	zpdes	9	2.625	1.400	0.467	0.533
S4	baseline	11	1.852	1.442	0.435	0.779
	zpdes	9	2.236	1.761	0.587	0.787
S5	baseline	11	1.724	1.526	0.460	0.885
	zpdes	9	1.778	1.713	0.571	0.963
S8	baseline	11	1.557	1.438	0.433	0.923
	zpdes	9	1.822	1.822	0.607	1.000

Assumption Checks

Test for Equality of Variances (Levene's)

	F	df1	df2	p
value_1	0.039	1	18	0.845
value_4	1.093	1	18	0.310
value_5	0.582	1	18	0.455
value_8	1.836	1	18	0.192